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## Pinion blade drive mechanism for a laparoscopic vessel dissector


### Abstract

A surgical instrument comprises a handle assembly including an actuator. An elongate shaft extends distally from the handle assembly and defines a longitudinal axis. A reciprocating member extends at least partially through the elongate shaft, and is mounted for longitudinal motion through the elongate shaft in response to manipulation of the actuator. A drive mechanism includes a first rotating component coupled to the actuator to induce rotational motion in the first rotating component. A second rotating component is coupled to the first rotating component such that rotational motion in the first rotating component induces rotational motion in the second rotating component. The second rotating component is coupled to the reciprocating member such that rotational motion of the second rotating component induces longitudinal motion in the reciprocating member.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/390,091, filed on Apr. 22, 2019, now U.S. Pat. No. 11,576,715, which is a continuation of U.S. patent application Ser. No. 12/773,644, filed on May 4, 2010, now U.S. Pat. No. 10,265,118.

INTRODUCTION
(1) The present disclosure relates to an apparatus for joining and transecting tissue. In particular, the disclosure relates to an apparatus having a drive mechanism for advancing a blade through tissue.
BACKGROUND
(2) Instruments such as electrosurgical forceps are commonly used in open and endoscopic surgical procedures to coagulate, cauterize and seal tissue. Such forceps typically include a pair of jaws that can be controlled by a surgeon to grasp targeted tissue, such as, e.g., a blood vessel. The jaws may be approximated to apply a mechanical clamping force to the tissue, and are associated with at least one electrode to permit the delivery of electrosurgical energy to the tissue. The combination of the

mechanical clamping force and the electrosurgical energy has been demonstrated to join adjacent layers of tissue captured between the jaws. When the adjacent layers of tissue include the walls of a blood vessel, sealing the tissue may result in hemostasis, which may facilitate the transection of the sealed tissue. To transect the tissue, an operator may squeeze a trigger or manipulate a similar actuator to advance a sharpened blade distally through a channel defined in the jaws. Since it is generally not necessary to release the tissue captured between the jaws before the blade is actuated, an accurate cut may be formed that extends only through tissue that has been properly sealed. A detailed discussion of the use of an electrosurgical forceps may be found in U.S. Pat. No. 7,255,697 to Dycus et al.

(3) Certain surgical procedures may be performed more quickly and accurately with an electrosurgical forceps having relatively longer jaws than one having shorter jaws. To this end, electrosurgical forceps have become available with jaws 60 mm in length or more. Longer jaws, however, may tend to present difficulties in transecting the sealed tissue. Since longer jaws are associated with greater distances a blade must traverse to fully transect tissue captured between the jaws, a longer actuation stroke may be required by an operator. A longer stroke may prove to be awkward or cumbersome for an operator.

SUMMARY

(4) The present disclosure describes a surgical instrument including a handle assembly having an actuator mounted for manipulation through an actuation stroke. An elongate shaft extends distally from the handle assembly and defines a longitudinal axis. A reciprocating member extends at least partially through the elongate shaft and is mounted for longitudinal motion through the elongate shaft. A drive mechanism is included for inducing longitudinal motion in the reciprocating member in response to manipulation of the actuator through the actuation stroke. The drive mechanism includes a first rotating component defining a first circumference, wherein the first rotating component is coupled to the actuator about the first circumference such that the manipulation of the actuator through the actuation stroke induces rotational motion in the first rotating component. The drive mechanism also includes a second rotating component defining a second circumference wherein the second rotating component is coupled to the first rotating component such that rotational motion in the first rotating component induces rotational motion in the second rotating component. The second rotating component is further coupled to the reciprocating member about the second circumference such that rotational motion of the second rotating component induces longitudinal motion in the reciprocating member. The second circumference is greater than the first circumference.

(5) The first rotating component may include a pinion gear defining a plurality of discrete teeth engaged with the actuator. The actuator may include a trigger mounted for pivoting through an actuation stroke to define an angle. Alternatively, the first rotating component may include a pulley, which is coupled to the actuator by a belt, and the actuator may include a drive wheel mounted for rotational motion.

(6) The second rotating component may include a gear having a plurality of discrete teeth engaged with a rack also having a plurality of discrete teeth and being mounted for longitudinal motion. Each of the plurality of discrete teeth of the rack may encircle a circumferential surface of the rack, and the rack may be mounted for rotational motion about the longitudinal axis.

(7) The surgical instrument includes an end effector extending distally from the elongate shaft. The end effector may include a pair of jaw members wherein at least one jaw member is configured to move between an open position substantially spaced from the other of the pair of jaw members and a closed position wherein the jaw members are closer together. The reciprocating member may be coupled to a knife adjacent a distal end of the reciprocating member, and the knife may be selectively extendable through a knife channel defined in the jaw members. The jaw members may include at least one electrode connectable to a source of electrosurgical energy.

(8) According to another aspect of the disclosure, an electrosurgical instrument includes a handle

assembly including an actuator mounted for manipulation through an actuation stroke. An elongate shaft extends distally from the handle assembly and defines a longitudinal axis. An end effector extends distally from the elongate shaft. The end effector includes a pair of jaw members wherein at least one jaw member is configured to move between an open position substantially spaced from the other of the pair of jaw members and a closed position wherein the jaw members are closer together. The pair of jaw members includes at least one electrode connectable to a source of electrosurgical energy. A reciprocating member extends at least partially through the elongate shaft and is mounted for longitudinal motion through the elongate shaft. A knife is coupled to the reciprocating member and is extendable into the end effector. A drive mechanism for inducing longitudinal motion in the reciprocating member in response to manipulation of the actuator through the actuation stroke includes a gear set having a first gear engaging the actuator and a second gear engaging the reciprocating member. The gear set defines a gear ratio less than one. The gear ratio may be about 1:2.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present disclosure and, together with the detailed description of the embodiments given below, serve to explain the principles of the disclosure.

(2) FIG. **1** is a perspective view of a surgical instrument in accordance with an embodiment of the present disclosure;

(3) FIG. **2**A is a partial, cross-sectional view of a handle assembly of the instrument of FIG. **1** depicting a trigger in an un-actuated position for maintaining a blade in a retracted position;

(4) FIG. **2**B is a partial, cross-sectional view of the handle assembly of FIG. **2**A depicting the trigger in an actuated position for maintaining a blade in an advanced position;

(5) FIG. **3**A is a partial, cross-sectional view of an end effector of the instrument of FIG. **1** depicting the blade in the retracted position;

(6) FIG. **3**B is a partial, cross-sectional view of the end effector of FIG. **4** depicting the blade in the advanced position;

(7) FIG. **4** is a schematic view of a pinion drive mechanism operated through an actuation stroke defining an angle;

(8) FIG. **5** is a schematic view of a pin-and-slot drive mechanism operated through an actuation stroke defining the same angle represented in FIG. **4**; and

(9) FIG. **6** is a partial, cross-sectional view of a handle assembly depicting an alternate embodiment of a drive mechanism.

DETAILED DESCRIPTION

(10) Referring initially to FIG. **1**, an embodiment of an electrosurgical instrument is depicted generally as **10**. The instrument **10** includes a handle assembly **12** for remotely controlling an end effector **14** through an elongate shaft **16**. Although this configuration is typically associated with instruments for use in endoscopic surgical procedures, various aspects of the present disclosure may be practiced in connection with traditional open procedures as well.

(11) Handle assembly **12** is coupled to an electrosurgical cable **20**, which may be used to connect the instrument **10** to a source of electrosurgical energy. The cable **20** extends to connector **22** including prong members **22***a* and **22***b* that are dimensioned to mechanically and electrically connect the instrument **10** to an electrosurgical generator (not shown). Each of the two prong members **22***a* and **22***b* may be associated with an opposite electrical terminal or potential (supplied by the generator) such that bipolar energy may be conducted through the cable **20**, and to the end effector **14**. Alternatively, the instrument **10** may include a battery and/or a generator (not shown)

disposed onboard the instrument **10** such that the instrument may operate as a self-contained unit.

(12) To control the end effector **14**, the handle assembly **12** includes a stationary handle **24** and movable handle **26**. The movable handle **26** may be separated and approximated relative to the stationary handle **24** to respectively open and close the end effector **14**. In FIG. **1**, the end effector **14** is depicted in an open configuration wherein upper and lower jaw members **32** and **34** are separated from one another such that tissue may be received therebetween. The jaw members **32**, **34** are pivotally coupled to the elongate shaft, and thus may be moved to a closed configuration (See FIG. **3**A) wherein the jaw members **32**, **34** are closer together for clamping the tissue. Any known mechanism may be incorporated into the instrument **10** to operatively couple the jaw members **32**, **34** to the handle assembly **12** including those described in U.S. Pat. No. 7,101,371 to Dycus et al. Handle assembly **12** also includes a rotation knob **36**, which may be manipulated to rotate the end effector **14** about a longitudinal axis A-A defined by the elongate shaft **16**, and a trigger **38** which is operable to advance a blade or knife **40** (see FIG. **3**A) through end effector **14** as discussed below with reference to FIGS. **2**A through **3**B.

(13) Referring now to FIG. **2**A, a clamping mechanism **44** for closing the jaw members **32**, **34** includes a drive rod **46** extending through the elongate shaft **16**. The drive rod **46** is coupled at a proximal end to a coupling member **50**, which is slidably disposed within the stationary handle **24**. The coupling member **50** is operatively coupled to the movable handle **26** such that as the movable handle **26** is approximated with the stationary handle **24** in the direction of arrow "A," the coupling member **50** is driven distally in the direction of arrow "a." The drive rod **46** is, in turn, also driven in the direction of arrow "a" by the coupling member **50**. Such longitudinal motion in drive rod **46** may effect pivotal motion in the jaw members **32**, **34** as described below with reference to FIG. **3**A.

(14) A pinion drive mechanism **54** for advancing the knife **40** is also disposed in the handle assembly **24**. The pinion drive mechanism **54** is operable by trigger **38**, which is pivotally mounted to stationary handle **24** about pivot pin **56**. Trigger **38** includes a grip **58** protruding to an exterior of the stationary handle **24** such that the grip **58** is accessible by an operator while gripping handle assembly **12**. The grip **58** is disposed at a first distance "d**1**" from pivot pin **56**. Opposite the grip **58**, on an interior of the stationary handle **24**, the trigger **38** includes a curved driving face **60**. The driving face **60** is centered about pivot pin **56** at second distance "d**2**," and includes a plurality of discreet teeth **60***t*. The trigger **38** defines a simple lever mechanism such that, for a given force applied at the grip **58**, a driving force delivered at the driving face **60** may be varied by varying the distances d**1** and d**2** from the pivot pin **56**.

(15) The teeth **60***t* of trigger **38** engage a circular gear set **64** that is rotatably coupled to the stationary handle **24**. The gear set **64** includes a smaller pinion gear **66** and a larger intermediate gear **68**. Although the ratio between the diameters of pinion gear **66** and intermediate gear **68** may vary, as depicted the smaller pinion gear **66** has a diameter approximately half the diameter of the larger intermediate gear **68**, and thus exhibits a circumference about half the circumference of the intermediate gear **68**. The gear set **64** defines a gear ratio of about 1:2, and thus, a tooth **68***t* on intermediate gear **68** travels approximately twice the distance as a tooth **66***t* on the pinion gear per rotation of the gear set **64**. The teeth **66***t* of the pinion gear **66** engage the teeth **60***t* of the trigger driving face **60** while the teeth **68***t* of the intermediate gear **68** engage a cylindrical rack **70**.

(16) The cylindrical rack **70** is movably disposed about the drive rod **46**, such that the rack **70** may translate longitudinally within the handle assembly **12**. The rack **70** includes a plurality of circumferential teeth **70***t* encircling an outer surface thereof. The circumferential teeth **70***t* permit the rack **70** to maintain engagement with intermediate gear **68** regardless of the radial orientation of the rack **70** about the longitudinal axis A-A. Thus, the rack **70** is configured to rotate along with rotation knob **36**. A distal end of the cylindrical rack **70** fixedly engages a knife tube **72** such that both rotational and longitudinal motion may be transferred between the knife tube **72** and the cylindrical rack **70**. The knife tube **72** extends through the elongate shaft **16** such that the knife tube **72** may be coupled to knife **40** (FIG. **3**A) as described in greater detail below.

(17) In use, the trigger **38** may be moved from an un-actuated position, as depicted in FIG. **2**A, in the direction of arrow "B", to an actuated position, as depicted in FIG. **2**B to advance the knife tube **72** distally. As the trigger **38** rotates about pivot pin **56**, the teeth **60***t* engage teeth **66***t* to drive the gear set **64** in the direction of arrow "b." The teeth **68***t* of the gear set **64**, in turn, drives the cylindrical rack **70** and knife tube **72** distally in the direction of arrow "bb." The knife tube **72** may be retracted by returning the trigger **38** to the un-actuated position. The pinion drive mechanism **54** exhibits a reduction in force applied at the knife tube **72** as compared to a force applied to the trigger **38**. Where greater forces are required, the trigger **72** may be modified by appropriately altering the distances "d**1**" and "d**2**."
(18) Referring now to FIGS. **3**A and **3**B, the knife tube **72** is coupled to knife **40** proximate a distal end of the knife tube **72**. The knife **40** is bifurcated to define a central elongate slot **74** therein. The elongate slot **74** permits longitudinal reciprocation of the drive rod **46** therethrough in response to manipulation of the movable handle **26** (FIG. **2**A). The drive rod **46** is coupled to jaw members **32**, **34** by a drive pin **76** extending through angled slots **78** defined in the jaw members **32**, **34**. The drive pin **76** reciprocates longitudinally with the drive rod **46** such that the drive pin **76** engages the angled slots **78** to cause jaw members **32**, **34** to pivot about pivot pin **80**, and thus moves the jaw members **32**, **34** between open and closed positions.
(19) Reciprocation of the knife tube **72** in response to manipulation of the trigger **38**, induces reciprocation of the knife **40** between a retracted position as depicted in FIG. **3**A, and an advanced position as depicted in FIG. **3**B. The retracted position of the knife **40** corresponds to the un-actuated position of trigger **38** (FIG. **2**A) and the advanced position of the knife **40** corresponds to the actuated position of the trigger (FIG. **2**B). In the advanced position, a sharpened cutting edge **82** of the knife **40** extends into a knife channel **84** defined in the jaw members **32**, **34**. The knife channel **84** extends between laterally disposed electrodes **86** on the jaw members **32**, **34**, which are configured to electrosurgically seal or otherwise treat tissue. The knife **40** may thus be moved from the retracted position wherein the cutting edge **82** is proximal to the electrodes **86** to the advanced position while transecting tissue captured between the electrodes **86**. The length of the jaws **32**, **34** defines the knife stroke "S," or the distance the knife **40** must travel between the retracted and advanced positions, to fully transect tissues captured between the electrodes **86**. The knife stroke "S," corresponds to the distance the rack **70** travels as the trigger **38** is moved between the un-actuated and actuated positions (see FIGS. **2**A and **2**B).
(20) For a given actuation stroke, a pinion drive mechanism **54***a* as depicted in FIG. **4** may offer a greater knife stroke "S" than a pin-and-slot drive mechanism **54***b* as depicted in FIG. **5**. The pinion drive mechanism **54***a* includes a trigger **28***a*, which is pivotable through an actuation stroke defining an angle "α". Movement of the trigger **28***a* through the actuation stroke rotates a gear set **64***a*, which in turn drives the rack **70***a* in a longitudinal direction as described above with reference to FIGS. **2**A and **2**B. The rack **70***a*, and thus a knife tube **72***a*, is translated through knife stroke "S."
(21) The pin-and-slot mechanism **54***b* includes a trigger **28***b*, which is pivotable through an actuation stroke defining the same angle "α". The trigger **28***b* includes a slot **88**, which engages a pin **90** of a translation block **70***b*. The translation block **70***b* is driven in a longitudinal direction as the trigger **28***b* pivots. Movement of the trigger **28***b* though the actuation stroke, however, translates the block **70***b*, by only a reduced knife stroke "s."
(22) For components sized and positioned suitably for use in a surgical instrument, the knife stroke "S" and the reduced knife stroke "s" have been empirically determined for an actuation stroke defining an angle "α" of 47.97°. The pinion drive mechanism **54***a* yielded a knife stroke "S" of 54.6 mm while the pin-and-slot drive mechanism yielded a reduced knife stroke "s" of 19.8 mm. A pinion drive mechanism **54***a* may achieve such a knife stroke "S" when an intermediate gear **68***a* exhibits a diameter approximately twice the diameter of a pinion gear **66***a*.
(23) Referring now to FIG. **6**, an alternate embodiment of a drive mechanism **102** is housed in a handle assembly **112**. Drive mechanism **102** includes a drive wheel **114** rotatably mounted to a

stationary handle **118**. The drive wheel **114** protrudes to an exterior of the stationary handle **118** such that the drive wheel **114** is accessible by an operator while gripping handle assembly **112**. A drive pulley **122** is mounted on the drive wheel **114** to such that the drive pulley **122** rotates along with the drive wheel **114**. The drive pulley **122** engages a belt **126**, which extends to a follower pulley **128**. The belt **126** assumes a “figure-eight” configuration such that follower pulley **128** may rotate in an opposite direction than the drive pulley **122**. A standard loop configuration is also contemplated for the belt **126** wherein the follower pulley **128** and drive pulley **122** rotate in the same direction. The follower pulley **128** is mounted on an intermediate gear **130** such that the intermediate gear **130** rotates along with the follower pulley **128**. The intermediate gear **130** drives rack **70**, which advances knife tube **72** as described above with reference to FIG. **2**A.
(24) In use, an operator may rotate the drive wheel **114** to induce longitudinal motion in the knife tube **72**. As the operator rotates drive wheel **114**, drive pulley **122** rotates one revolution for each revolution of the drive wheel **114**. Drive pulley **122**, in turn, drives follower pulley **128**. The follower pulley **128** may define a smaller circumference than the drive pulley **122** such the follower pulley **128** rotates more than one revolution for each revolution of the drive pulley **122** and drive wheel **114**. The intermediate gear **130** rotates one revolution for each revolution of the follower pulley **128**. Since the intermediate gear **130** defines a larger circumference than the follower pulley **128**, the intermediate gear **130** may drive the rack **70** and knife tube **72** a greater longitudinal distance than a direct connection between the rack and drive wheel **114**.
(25) Although the foregoing disclosure has been described in some detail by way of illustration and example, for purposes of clarity or understanding, it will be obvious that certain changes and modifications may be practiced within the scope of the appended claims.

## Claims

1. A surgical instrument, comprising: a housing; an elongate shaft extending distally from the housing; a pair of jaw members disposed at a distal end of the elongate shaft and configured to grasp tissue; a rack disposed within the housing and configured to move along a longitudinal axis, the rack having a plurality of teeth; a drive rod disposed through the rack and configured to move at least one jaw member of the pair of jaw members; a first gear disposed within the housing and out of contact with the rack; and a second gear concentric with the first gear and having a plurality of teeth configured to engage the plurality of teeth of the rack, wherein rotation of the first gear induces rotation of the second gear to cause movement of the rack along the longitudinal axis.
2. The surgical instrument according to claim 1, wherein the second gear is fixed relative to the first gear.
3. The surgical instrument according to claim 1, further comprising a trigger having a first portion disposed exterior to the housing and a second portion disposed within the housing, wherein the second portion is in direct contact with the first gear and configured to rotate within the housing to cause rotation of the first gear.
4. The surgical instrument according to claim 3, wherein the second portion of the trigger defines a plurality of teeth configured to engage a plurality of teeth defined by the first gear.
5. The surgical instrument according to claim 3, wherein the trigger is configured to rotate about a pivot pin disposed within the housing.
6. The surgical instrument according to claim 1, further comprising a knife tube coupled to the rack and configured to move along the longitudinal axis in response to movement of the rack along the longitudinal axis.
7. The surgical instrument according to claim 6, wherein the drive rod is configured to move longitudinally within the knife tube.
8. The surgical instrument according to claim 6, further comprising a knife coupled to a distal end portion of the knife tube and configured to move along the longitudinal axis.

9. The surgical instrument according to claim 6, wherein the knife tube is disposed distal to the rack.
10. The electrosurgical instrument according to claim 1, wherein the drive rod is configured to move relative to the rack.
11. The electrosurgical instrument according to claim 1, wherein a proximal end portion of the drive rod is disposed proximal to a proximal end of the rack.
12. A surgical instrument, comprising: a housing; a trigger coupled to the housing and having a first portion disposed exterior to the housing and a second portion disposed within the housing, the second portion having a plurality of teeth; a rack disposed within the housing configured to move along a longitudinal axis; a drive rod disposed through the rack and configured to move along the longitudinal axis; a first gear having a plurality of teeth in direct contact with the plurality of teeth of the second portion of the trigger such that rotation of the trigger rotates the first gear; and a second gear coupled to the rack, wherein rotation of the first gear induces rotation of the second gear to move the rack along the longitudinal axis.
13. The surgical instrument according to claim 12, wherein the first and second gears are concentric.
14. The surgical instrument according to claim 12, further comprising a knife tube coupled to the rack and configured to move along the longitudinal axis in response to movement of the rack.
15. The surgical instrument according to claim 14, wherein the drive rod is configured to move longitudinally within the knife tube.
16. The surgical instrument according to claim 12, further comprising a handle configured to move relative to the housing to cause movement of the drive rod along the longitudinal axis.
17. The surgical instrument according to claim 16, wherein the trigger is configured to rotate relative to the handle and the housing to rotate the first gear.
18. The surgical instrument according to claim 12, wherein the second gear defines a plurality of teeth configured to engage a plurality of teeth defined by the rack.
19. A surgical instrument, comprising: a housing; a rack disposed within the housing and configured to move along a longitudinal axis; a drive rod disposed through the rack and configured to move relative to the rack along the longitudinal axis; a handle configured to rotate relative to the housing to move the drive rod along the longitudinal axis; a trigger configured to rotate relative to the handle and the housing; a first gear coupled to the trigger such that rotation of the trigger rotates the first gear; and a second gear coupled to the first gear and the rack, wherein rotation of the first gear induces rotation of the second gear to move the rack along the longitudinal axis.
20. The surgical instrument according to claim 19, wherein the first and second gears are concentric.